

Deep Learning for Indoor Scene Style Classification

**A Project Report
Scientific Computing Department
Neural Network - Fall 2025**

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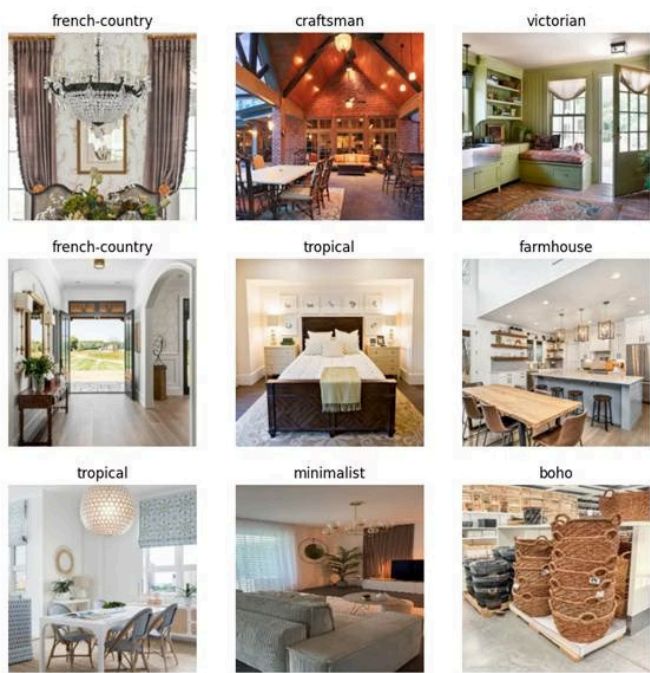
Project Team

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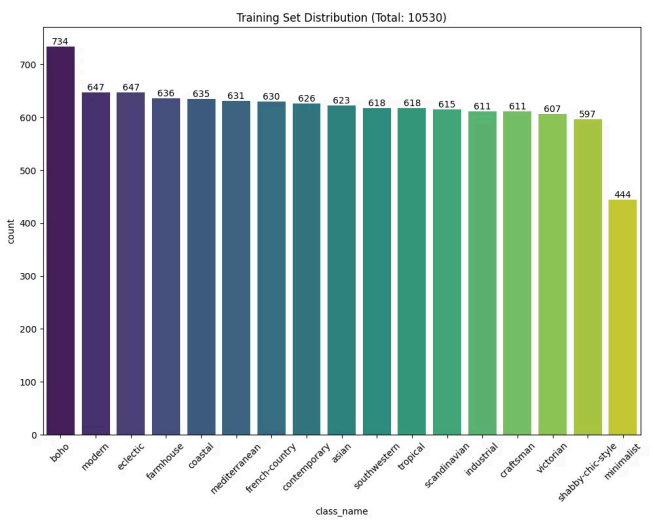
Project Overview

The project focuses on training deep learning models to classify indoor scenes into various design styles based on visual characteristics. Utilizing a dataset of style-labeled room images, the model aims to automatically identify and distinguish interior design styles, ensuring consistency and accurate recognition across diverse environments.

Data Samples



Classes Distribution



Preprocessing: Grouping and Splitting

Grouping 17 Classes into 4 Super Groups

- **Group 0: "Sleek, Hard, Geometric"** (modern, contemporary, minimalist, industrial, asian)
- **Group 1: "Soft, White, Airy"** (scandinavian, coastal, shabby-chic-style)
- **Group 2: "Historic, Ornate, Luxury"** (farmhouse, french-country, craftsman, victorian, mediterranean)
- **Group 3: "Organic, Textural, Busy"** (boho, eclectic, tropical, southwestern)

Train/Validation Split & Class Weighting

Data is split into 80% training (10,530 images) and 20% validation (2,633 images) using stratification. Class weighting is applied to handle imbalance, with calculated weights for both super groups and individual styles to ensure fair representation during training.

Advanced Data Augmentation Strategy

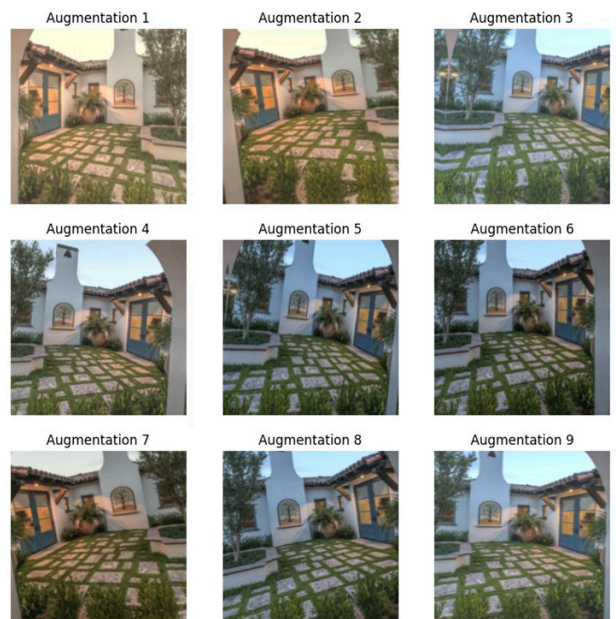
To enhance model generalization and prevent overfitting, a dynamic data augmentation pipeline was implemented using TensorFlow/Keras. This pipeline introduces synthetic variability to training data at runtime, ensuring the model learns features invariant to camera framing, orientation, and lighting conditions.

Geometric Transformations

- Random Flip (horizontal)
- Random Translation (up to 10%)
- Random Rotation (approx. $\pm 10^\circ$)
- Random Zoom (up to 20%)

Photometric Transformations

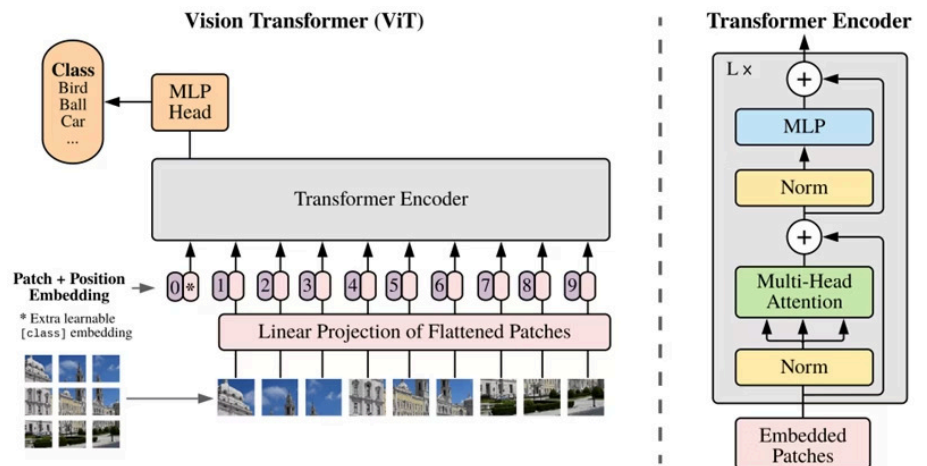
- Random Brightness & Saturation ($\pm 20\%$)
- **Planckian Jitter (Custom Layer):** Simulates realistic lighting temperature shifts, approximating "warm" vs. "cool" conditions without distorting semantic content.



Models Trained: CLIP Architectures

The project leveraged transfer learning with state-of-the-art architectures, including Vision Transformers (ViTs) from the CLIP family, as feature extractors with custom classification heads.

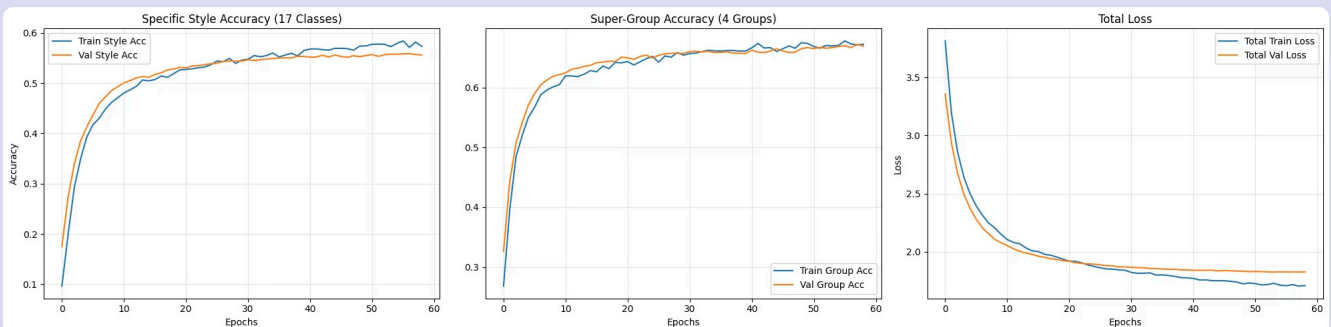
CLIP models, pre-trained on text-image pairs, offer superior semantic understanding compared to purely visual CNNs.



CLIP ViT-Large (Patch 14)

A larger variant trained on LAION-2B, using 14x14 patches for finer granularity and a 768-dimensional space.

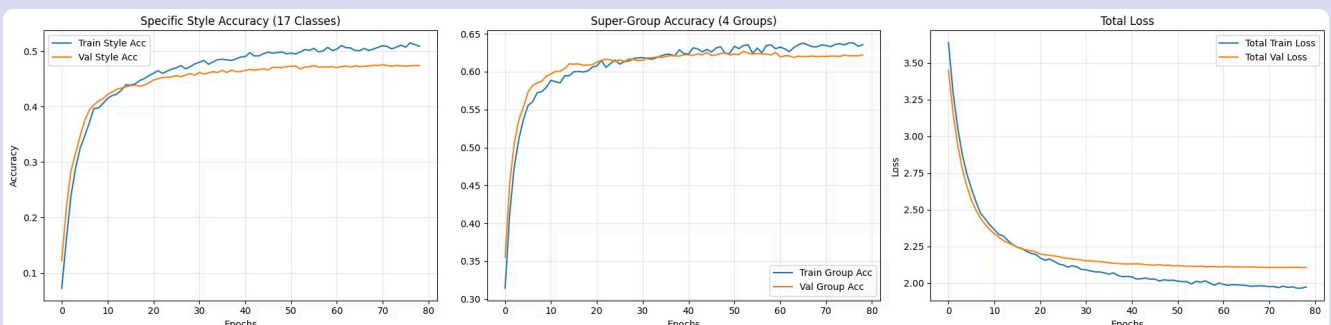
- Total params: 304,250,901, Trainable params: 1,071,125, Accuracy: 0.56



CLIP ViT-Base (Patch 32)

Processes images as 32x32 patches, projecting into a 512-dimensional embedding space.

- Total params: 88,264,981, Trainable params: 808,981, Accuracy: 0.47

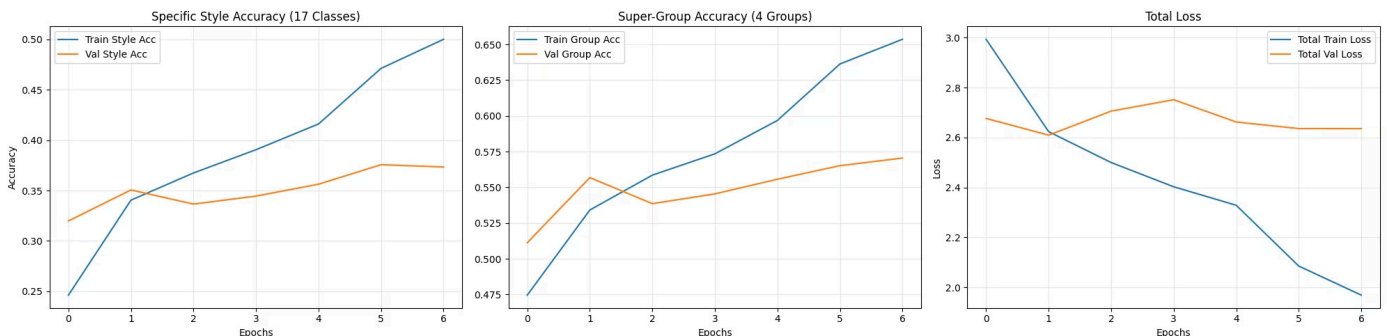
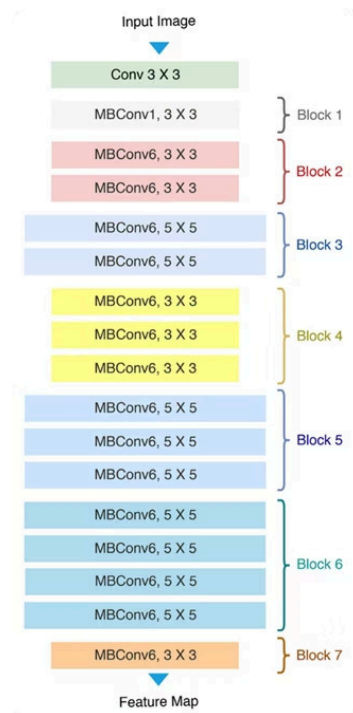


EfficientNet Family

Custom EfficientNet architectures were implemented, utilizing MBConv blocks with squeeze-and-excitation optimization and ImageNet pre-trained weights.

EfficientNet B0

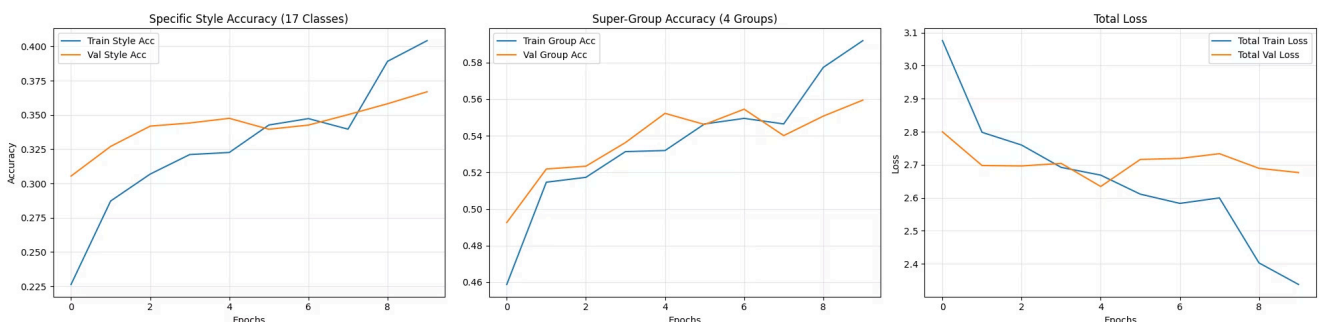
Baseline network configured for 224x224 input resolution.



EfficientNet B3

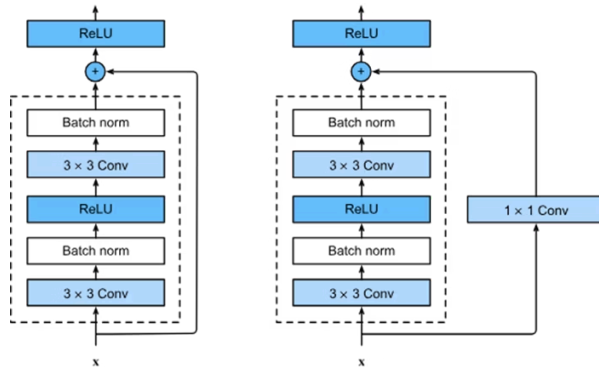
Larger variant with increased width/depth, configured for 300x300 input to capture detailed spatial features.

- Total params: 11,581,245, Trainable params: 797,717, Accuracy: 0.35

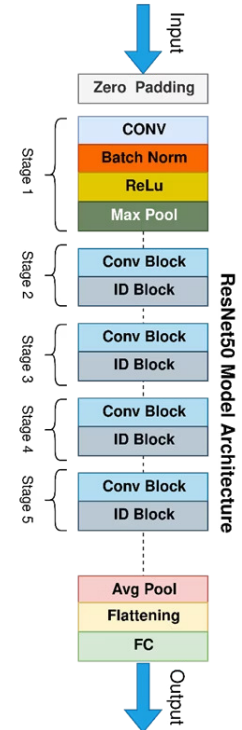
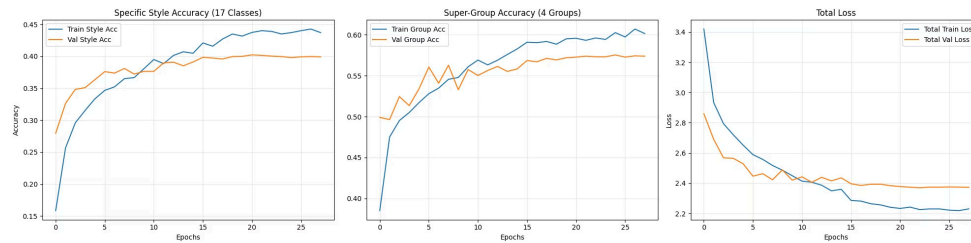


ResNet50

A standard 50-layer Residual Network, used as a baseline CNN for robust hierarchical feature extraction.



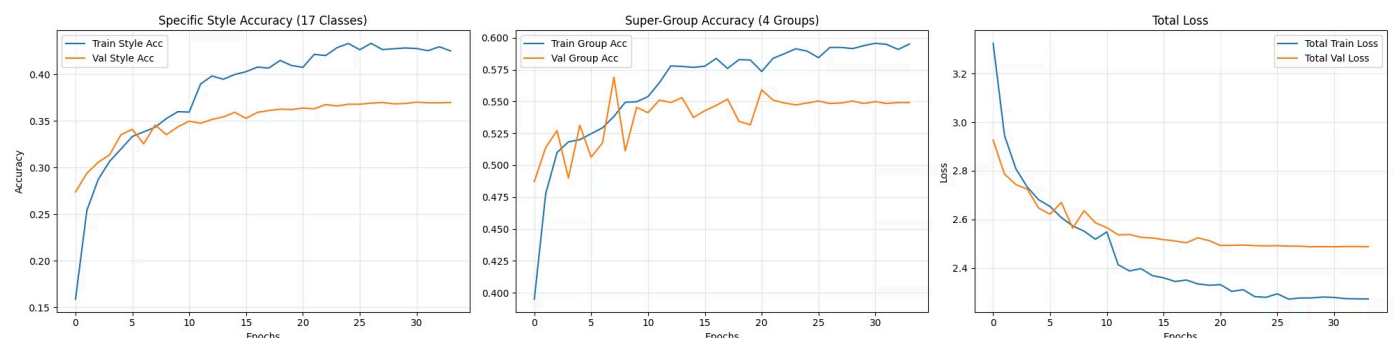
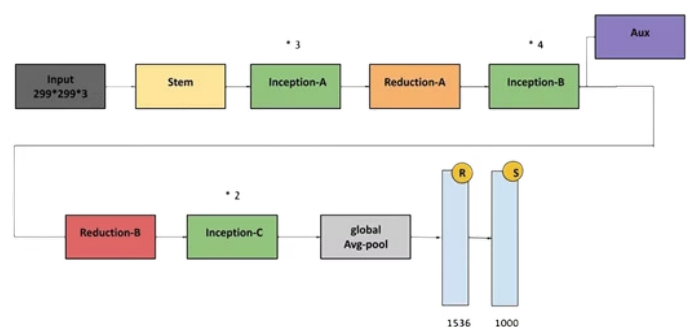
Total params: 24,647,573, Trainable params: 1,059,861, Accuracy: 0.40



InceptionV3

Known for factorized convolutions and aggressive regularization, it excels at capturing multi-scale features for complex scene patterns.

Total params: 22,862,645, Trainable params: 1,059,861, Accuracy: 0.37



Training Dynamics and Optimizers

All models utilized Transfer Learning, replacing original classification heads with custom layers. This multi-model approach enhances robustness by aggregating predictions from diverse architectures.

Early Stopping & ReduceLROnPlateau

Prevented overfitting and saved resources by stopping training when validation loss stalled and dynamically lowering the learning rate.

Model Checkpointing

Ensured only the best model version (lowest validation loss) was saved, not just the final epoch.

Optimizers

SGD with momentum for ResNet50/InceptionV3, and RMSprop/Adam for EfficientNet, based on architectural recommendations for stability.

Preprocessing Analysis

Model	Input Size	Color	Value Range	Logic
CLIP VIT (Patch32 / L14)	224x224	RGB	~ -1.8 to +2.2	Normalize (Mean/Std)
ResNet50	224x224	BGR	~ -120 to +150	preprocess_resnet_input (Mean Subtraction)
EfficientNet B0/B3	224/300	RGB	~ -2 to +2	preprocess_efficientnet_input (Std Dev)
InceptionV3	299x299	RGB	-1 to +1	preprocess_inception_input (Linear Scale)

Accuracy Analysis

Model Architecture	Parameters	Epochs	Training Accuracy	Val Accuracy
CLIP (ViT patch32)	~88M	79	0.50	0.47
CLIP (ViT-L/14)	~304M	59	0.57	0.55
ResNet50	~24M	28	0.42	0.39
EfficientNetB0	~4M	7	0.48	0.37
EfficientNetB3	~11M	10	0.40	0.36
InceptionV3	~22M	34	0.42	0.37

Analysis Note: **CLIP** (despite its size) likely offers the best semantic understanding because it was pre-trained on text-image pairs, whereas CNNs like **ResNet** focus purely on visual patterns.

Technical Challenges & Solutions

- 1

Computational Optimization
Mixed precision training (float16) and data prefetching/parallelism (tf.data.AUTOTUNE) were used to handle large batch sizes (256) and complex models, reducing memory and speeding up training on GPUs.
- 2

Robust Data Handling
A custom PIL-based image loader was implemented to handle corrupt images, returning a black image for unreadable files instead of crashing the pipeline, ensuring complete testing.
- 3

Evaluation Metrics
Beyond standard accuracy, Top-3 Accuracy was monitored, considering a prediction correct if the true style was among the top 3. Per-head loss for both 17 styles and 4 super-groups ensured learning of granular and high-level features.